

25 Plane polarised light of intensity I is incident normally on a set of polarising filters.

The transmission axis of the first filter is at 30° to the plane of polarisation of the incident light.

The transmission axis of each successive filter is at 30° to the transmission axis of the previous filter.

The intensity of the light emerging from the final filter is $0.24I$.

How many polarising filters are in the set?

A 3

B 5

C 6

D 10